

Digital Record Project – Comprehensive Professional Training Plan

- Duration: 8 Weeks
- Target Audience: Summer Internship Students (Software Development Track)

Introduction

The Digital Record project is a smart platform designed to analyze students' academic skills and generate personalized career path recommendations using AI.

Student data is read from a simulated JSON file (Mock API) and sent to OpenAI API for processing, after which results are displayed visually.

Training Objectives

1. Train students to develop a full-stack Laravel application with MySQL.
2. Understand and use AI models via OpenAI API.
3. Read, parse, and process JSON-based academic data.
4. Build intelligent dashboards to display analysis and suggestions.
5. Collaborate via GitHub using Git Flow methodology.
6. Write professional documentation, including Software Requirements Specification (SRS).

Key Features (Tasks)

1. Smart Student Dashboard

Displays key performance indicators and skill progress using visual charts.

Includes: progress bars, strongest/weakest skills, AI-driven insights.

- Technologies: Laravel + Chart.js
- DB Tables: student_skills, ai_analytics

2. Interactive PDF Career Profile

Allows students to download a customized PDF containing their academic summary, skill achievements, career paths, and AI suggestions.

- Technologies: Laravel + DomPDF
- Output: Professionally designed PDF template and download button

3. Peer Comparison Tool

Enables students to compare their progress with peers in the same department.

- Metrics: Skill averages, ranking, strengths per group
- Technologies: SQL aggregates + visual comparison interface

4. Smart Notifications Center

Automatically sends alerts to students such as:

- “You’ve improved your critical thinking skill!” or “Consider completing the recommended Python course.”
- Technologies: Laravel Notifications + notifications table

5. Optional: Chatbot Assistant

A smart assistant that interacts with the student, answers questions about their skills or future plans, and offers insights.

- Technologies: OpenAI Chat API (ChatCompletion)
- UI: Web-based chat window integrated in dashboard

8-Week Timeline

- Week 1: Laravel basics + project setup + GitHub and JSON structure
- Week 2: Data mapping + rendering from JSON
- Week 3: OpenAI API integration (POST and response handling)
- Week 4: Store responses in DB + model creation
- Week 5: Feature development begins (teams assigned)
- Week 6: Write and refine SRS + test integrated features
- Week 7: Improve user experience + final polish
- Week 8: Final documentation + SRS submission

System Requirements

Software:

- Laravel 10+
- MySQL / MariaDB
- HTML, CSS, JavaScript
- Node.js + NPM
- OpenAI API Key
- GitHub account

Hardware:

- Laptop/Desktop with minimum i5 processor
- 8GB RAM minimum
- Stable internet connection
- Code editor (e.g. VS Code or PhpStorm)

Helpful Resources

- Laravel Docs: <https://laravel.com/docs>
- OpenAI Docs: <https://platform.openai.com/docs>
- GitHub Learning Lab: <https://lab.github.com/>
- Working with JSON: <https://laravel-news.com/working-with-json>
- SRS writing:
https://www.tutorialspoint.com/software_engineering/software_requirements.htm
- Postman for API testing: <https://www.postman.com/>

Best Practices

- Use GitHub Projects to organize issues and milestones.
- Follow Git Flow: main, develop, feature/*
- Commit frequently with descriptive messages.
- Write unit/feature tests using Laravel testing tools.
- Each team must write and submit an SRS document for their feature.

Team Deliverables

1. Fully functional implementation of the assigned feature.

2. A completed Software Requirements Specification (SRS) document including:

- Overview and objective of the feature
- Functional and non-functional requirements
- UI mockups or diagrams
- APIs or DB schemas used